Grammatical Error Correction on C4_200M

Bag-of-Words → LSTM → Attention → Transformer

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CIE 555 — Neural Networks and Deep Learning

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The Problem: Grammatical Error Correction

Rewrite an ungrammatical sentence into its corrected form.

> **Input (corrupted):** "She go to school yesterday and learning many thing."
>
> **Output (corrected):** "She went to school yesterday and learned many things."

- Errors: tense, agreement, articles, plurals, punctuation, spelling.
- Output is a **whole new sentence** — a sequence-to-sequence task.

The Data: C4_200M Subset

Clean web text automatically corrupted into `(corrupted, clean)` pairs.

Split	Pairs
Train	850,000
Validation	50,000
Test	100,000
Total	**1,000,000**

- 16k Byte-Level BPE tokenizer.
- Max length 64 tokens.
- Same splits + tokenizer for every model — no leakage.

Four Models of Increasing Power

#	Model	What it adds
1	Bag-of-Words (TF-IDF)	Baseline — only **detects** errors
2	LSTM encoder–decoder	First corrector; fixed-size bottleneck
3	LSTM + Bahdanau attention	Decoder re-queries the full source
4	Transformer	Self-attention; no recurrence

Model 1 **detects**. Models 2–4 **correct**.

Model 1 — Bag-of-Words Baseline

- Each pair → two samples: corrupted = ungrammatical (0), clean = grammatical (1).
- Features: TF-IDF, unigrams + bigrams, 50k vocab (fit on train only).
- Classifiers: Logistic Regression & Linear SVM.

> **Result:** ~**62%** detection accuracy — barely above 50% chance. Surface word statistics cannot solve GEC → need seq2seq.

Model 2 — LSTM Encoder–Decoder

- BiLSTM encoder compresses the input into **one** 512-dim state.
- LSTM decoder generates the correction token by token.
- Teacher forcing 1.0→0.5; 19.55M parameters; 3 epochs.
- Test GLEU **0.213** (greedy) — the weakest corrector.

> **The bottleneck:** after the first step the decoder never "sees" the source again — everything is squeezed into 512 numbers.

Model 3 — LSTM + Bahdanau Attention

Keep **all** encoder states; the decoder attends to them every step.

`score(s,h) = v·tanh(W·s + W·h)  →  α = softmax(score)  →  context = Σ αᵢ·hᵢ`

- Removes the bottleneck; alignment weights are interpretable.
- Padding masked with $-\infty$ before softmax. 29.06M parameters.

> **The decisive jump:** GLEU 0.213 → **0.599** (+0.386). Attention matters most.

Model 4 — Transformer

- No recurrence — self-attention only.
- d_model 256, 8 heads, 3+3 layers, FFN 512.
- Sinusoidal positional encoding; causal mask in the decoder.
- Label smoothing 0.1; warmup learning rate; weight tying.

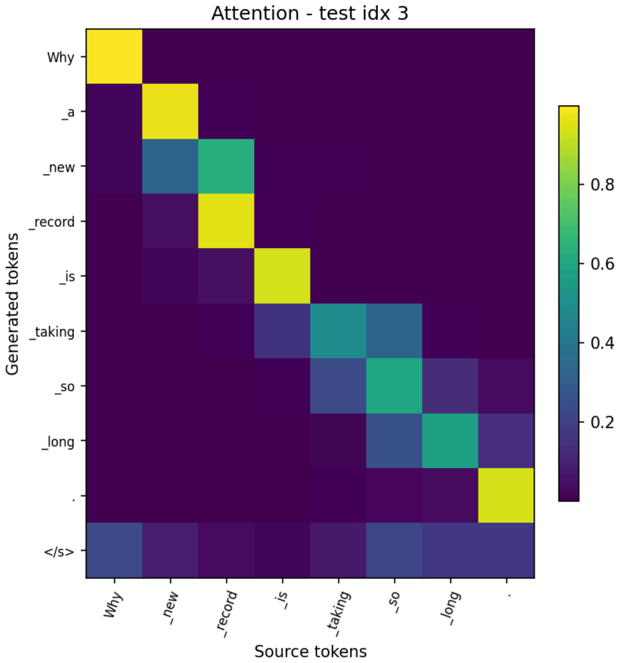
> **Smallest & strongest:** only **8.05M** parameters (3.6× fewer than Model 3). Best validation GLEU **0.613**.

Comparative Results — Test Set

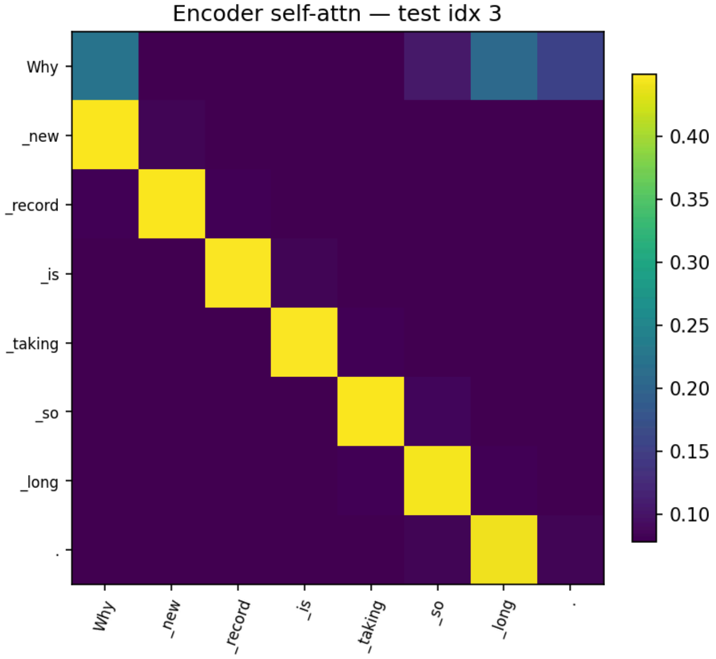
Model	Decoding	Params	EM	GLEU
LSTM (no attention)	greedy	19.55M	≈0.008	0.213
LSTM (no attention)	beam-4	19.55M	—	0.256
LSTM + Bahdanau	greedy	29.06M	0.024	0.599
LSTM + Bahdanau	beam-4	29.06M	0.026	**0.618**
Transformer	greedy	8.05M	0.022	**0.618**
Transformer	beam-4	8.05M	0.020	**0.618**

- Beam search helps the LSTM, not the (well-calibrated) Transformer.
- Transformer ties Model 3 on GLEU with 3.6× fewer parameters.

Attention Is Interpretable



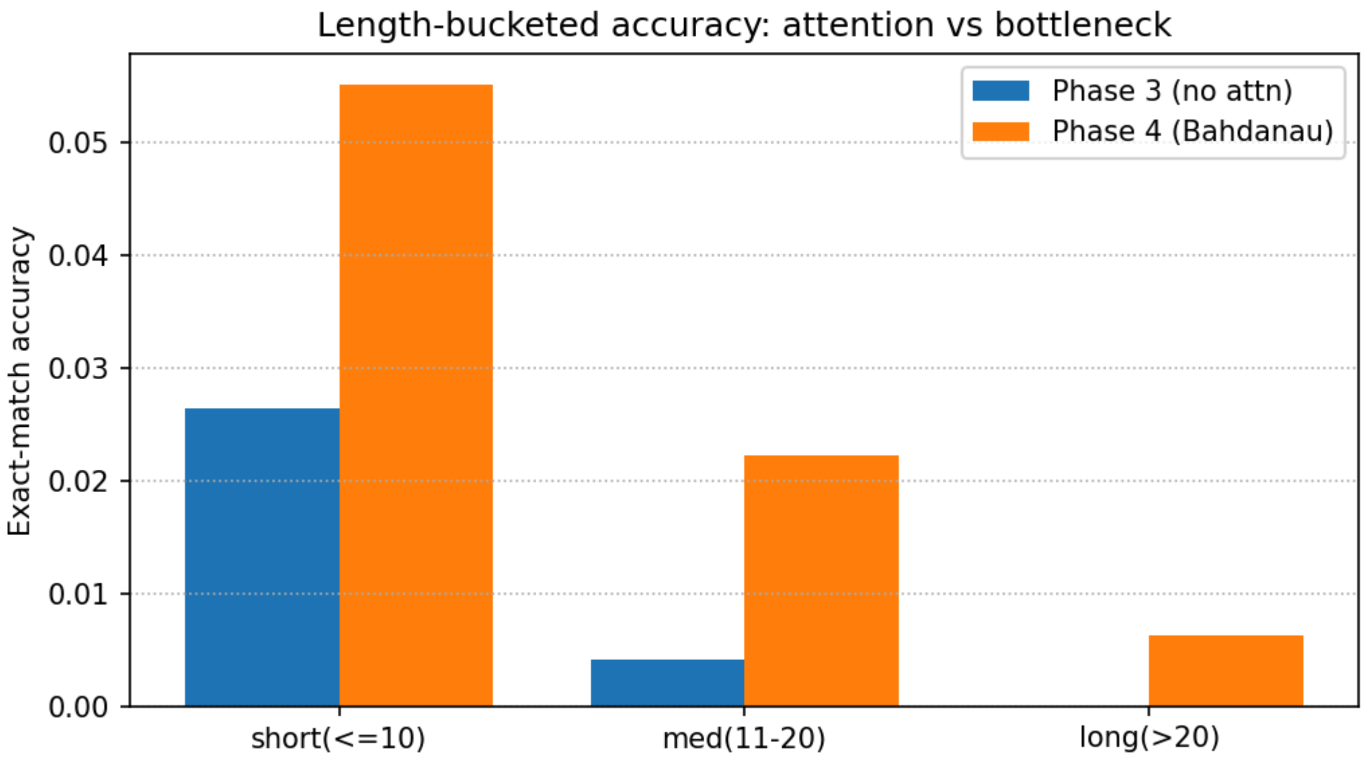
Bahdanau decoder alignment



Transformer encoder self-attention

Bright cells = which source token the model focuses on.

Behaviour by Sentence Length



All models do best on short sentences; the bottleneck LSTM degrades fastest as sentences get longer.

Grammar-Rule Analysis of Outputs

Learned well (local syntax):

- Subject–verb agreement: "What are this job" → "What is this job"
- Articles: "Why new record" → "Why the new record"
- Verb tense, punctuation, spacing.

Still hard:

- Long-range / semantic edits; rare-word spelling (hallucination).
- Dominant failure = under-correction — copying the source unchanged.

The Final Challenge — Dataset Quality

Why does **every** model score only ~2% exact match?

Reference quality is the ceiling — not model capacity

1. References are paraphrases: "informed"→"suggest", "startups"→"start-ups" — not grammar errors.

2. References can be noisy: target "no brain worky" — not a word.

3. Many valid corrections exist; exact match credits exactly one.

→ Report GLEU / F0.5, not EM. Cleaner data helps more than a bigger model.

Conclusion

- **BoW** detects grammaticality at ~62% — cannot correct.
- **LSTM** corrects but is limited by the bottleneck (GLEU 0.213).
- **Bahdanau attention** is the decisive jump (GLEU 0.599).
- **Transformer** ties on GLEU (0.618) with 3.6× fewer params, best validation GLEU (0.613).

> **Key insight:** the binding constraint is **dataset quality**, not architecture — synthetic C4_200M references are often paraphrases or noise.

Thank You

Questions?